



ADMINISTRATIVE INFORMATION

FMI
No: 25408A_B

FMI Title: SCSI Upgrade for CT/CTPET Systems with 2368838 Tower

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☐ Mandatory Safety	GEMS - AM:	Issue Date:
☒ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date:

- ☒ **Staged FMI** **Customer List** provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.
- ☐ **Parts** and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.
- ☐ **Parts Ordering:** GEMS - Americas, Contact your Region logistics coordinator
GEMS - Europe, Order FMI kits from GPO/EPL Buc
GEMS - Asia, Contact your region logistics coordinator
- ☐ **No Parts** are required to complete this FMI.
- ☒ **Batch with** **Batch with next service visit to the site to avoid travel charges.**
- To minimize costs, combine implementation of this FMI with the FMI's listed for all customers where the effectivity is the same (Please see your customer lists).
- ☐ **Non-Staged** **No Customer List is provided.** Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Online Center 800-321-7937 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: 33(0)1 30 83 10 00or ESC Fax: (33 1) 30 70 99 70	Please contact a support engineer in your region.

Special Instructions:

This FMI is only for systems with old 2368838 SCSI Tower. Not applicable for systems with new 5167410 SCSI Tower. *Applies to both CT and PET systems.*

If your site has an older version (2368838) SCSI Tower, you can use any of the 25408 instructions or kits.

Please perform this FMI in conjunction with the next service visit to the site to avoid travel charges.

Due dates have been extended to allow consolidation of trips.

Unit Tracked: Systems with GRE consoles with 2368838 Tower

FMI Instructions Part No.: 5317461-1EN, Rev 4

Do NOT leave unsecured on site

FMI Completion Certification Report

Mail To

GEMS - Americas	GEMS - Europe	
GE Medical Systems Attn: FMI Admin. W-523 P.O. Box 414 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: 33(0)1 30 83 10 00or ESC Fax: (33 1) 30 70 99 70	Send to your Country FMI or Service Administrator

FMI No.: 25408A_B

FMI Title: SCSI Upgrade for CT/CTPET systems with 2368838 Tower

Customer Full Name:

Number and Street Address:

City and Country:

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

Installer
Comments

Revision History

Revision	Date	Reason for Change
1	8/20/2008	Initial Release
2	9/30/2008	Changes to incorporate pilot feedback
3	10/17/2008	Change in travel and labor charges
4	11/18/2008	Combine 25408A and 25408B instructions to one document

Effectivity

The FMI effectivity includes GRE console systems with **old 2368838 SCSI Tower**.

This FMI applies to ALL CT and CTPET systems with GRE consoles.

Originally, FMI 25408 was broken into two distributions, FMI 25408A and 25408B for purposes of distributing the cost to the business. Both FMI's 25408A and 25408B had identical kits and instructions.

The documents and kits have now been combined into FMI 25408A/B. Therefore, any site with the old SCSI Tower 2368838 can use FMI 25408A, 25408B or 25408A/B regardless of which kit is *shipped*.

The FMI Console Model numbers tracked:

2337535

2337535-2

2366438 (H16 Linux)

2377708

2377708-4

2377708-7

2377708-10

2377708-13

2377708-14

2377708-15

2377708-20

2377708-21

2377708-22

2377708-30

2377708-31

2377708-32

FMI	FIELD MODIFICATION INSTRUCTION	FMI
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2377708-40		
2377708-41		
2377708-42		
2377708-43		
2377708-50		
2377708-51		
2377708-52		
2377708-56		
2377708-53		
2377708-120		
2377708-520		
2377708-530		
2377708-540		
2377708-550		
2377708-551		
2377708-552		
2377708-553		
2377708-554		
2377708-556		
5115335		
5115335-2		
5115335-10		
5115335-12		
5115335-800		
5111750-801		
5377708		
5377708-7		
5377708-9		
5377708-43		
5377708-120		
5377708-802		
5377708-520		
5377708-530		
5377708-560		

Purpose

This FMI address early life failures of the power supply used in the old SCSI tower, 2368838. Also, this FMI replaces the old DVD drive with the new Panasonic model SW-9576 that better fit the current CT options.

Prerequisites

None.

Furnished Materials

Kit Part Number: 5317461

Item #	Qty	Description	Part #
1	1	FMI Instructions	5317461-1EN
2	1	Peripheral Media Tower Chassis Assembly	5167410-10
3	1	5X DVD-RAM Multi Drive 4.7GB per side	2398863-2
4	1	Product Label	5323345
5	5	Jumper	5323050

Tools Required

- Standard and Phillips Screwdriver, Needle Nose Pliers
- ESD PPE / Equipment
- Dynema Gloves (Cut Resistant)

Personnel Requirements

Field Engineer: 1 Labor: 30 minutes

Please perform this FMI in conjunction with the next service visit to the site to avoid travel charges.

NOTICE



TO PREVENT DAMAGE TO COMPONENTS WITHIN THE SCSI TOWER, OBSERVE THE FOLLOWING ESD PRECAUTIONS:

- WORK ON A STATIC-FREE MAT.
- WEAR A STATIC STRAP TO ENSURE THAT ANY ACCUMULATED ELECTROSTATIC CHARGE IS DISCHARGED FROM YOUR BODY TO GROUND.
- CREATE A COMMON GROUND FOR THE EQUIPMENT YOU ARE WORKING ON BY CONNECTING THE STATIC-FREE MAT, STATIC STRAP AND PERIPHERAL UNITS TO THAT PIECE OF EQUIPMENT.

Procedure

1. Preliminary Checks

- 1.1 This FMI applies to only systems with the old SCSI Tower, model number 2368838. This FMI is not applicable to systems with new SCSI Tower, model number 5167410.

If your system already has the new 5167410 SCSI Tower, return the kit and close the FMI with 'code 2'.

The model number of the SCSI tower can be found in the rating plate located on the bottom of the SCSI Tower enclosure.

- 1.2 Most DVD drives used in 2368838 SCSI Towers are IDE drives with a SCSI Bridge Adapter Board (part# 2368521). It is possible, **but not likely** that there may be few sites with an older 2368838 SCSI tower and a DVD Drive that **does** not use the Bridge Adapter board. If you should experience that case, please return the FMI kit, order a new SCSI Tower, part # 5167410 for replacement in the next visit to the site.

To check the DVD drive prior to shutting down the system, use the following commands:

1. Open a Unix Shell (**or telnet window if done remotely**)
2. type **scsistat** (as in the example below)
3. Note the DVD-RAM listed in the printout.

Output will look similar to the example below. Note the CD-ROM DVD-RAM Model # in the first example is 9574. In the second example, it is a 2050.

The following are known CD-ROM, DVD-RAM Models. If you have a different model, you will have to physically check for the existence of the Bridge Adapter Board.

- Models D521, 9571, 9572, 9574, 9576 use the Adapter Board, proceed with FMI.
- Model 2050 does not use the Adapter Board. Return FMI kit and order part #5167410 **(there are likely very few, if any of these 2050 drives.)**

Example with SW9574S:

```
[insite@ct03 ~]$ scsistat
```

```
Device 0 03 Optical Device SONY SMO-F551-SD FW Rev: 1.05
```

```
Device 0 04 CD-ROM MATSHITA DVD-RAM SW-9574S FW Rev: AZX8
```

```
Device 2 00 Direct-Access SEAGATE ST336753LW FW Rev: HPS3
```

```
Device 3 01 Direct-Access SEAGATE ST373453LW FW Rev: HPS3
```

```
Device 3 02 Direct-Access SEAGATE ST373453LW FW Rev: HPS3
```

Example with DVD-RAM Hitachi GF-2050:

```
[insite@ct03 ~]$ scsistat
```

```
Device 0 03 Optical Device SONY SMO-F551-SD FW Rev: 1.04
```

```
Device 0 04 CD-ROM HITACHI DVD-RAM GF-2050 FW Rev: E372
```

```
Device 2 00 Direct-Access SEAGATE ST336753LW FW Rev: HPS2
```

```
Device 3 01 Direct-Access SEAGATE ST373453LW FW Rev: HPS2
```

```
Device 3 02 Direct-Access SEAGATE ST373453LW FW Rev: HPS2
```

2. Power Off Operator Console

- 2.1. Detach and remove the disks from both MOD and DVD drives
- 2.2. Select one of the following methods to Power Off the Operator Console:
 - If Applications are running, click on the [Shut Down] icon and select [**Shutdown**]
 - If Applications are down, open a Unix Shell using the Toolchest.

Type: {ctuser@hostname} **halt** [ENTER]

- 2.3. The Operator Console monitor will display a 'System Halted' message when it is acceptable to power off the Operator Console. Power off the Operator Console by using the front panel switch.

NOTE:

If console power is not turned off per above procedure, when SCSI unit is connected to the system after the upgrade, the system will not recognize SCSI devices and a reboot will be required.

3. Disconnect old SCSI Tower

- 3.1. Remove the power cord at the rear of the SCSI Tower chassis.
- 3.2. Remove the SCSI cable connection from the back of the SCSI Tower chassis being replaced.

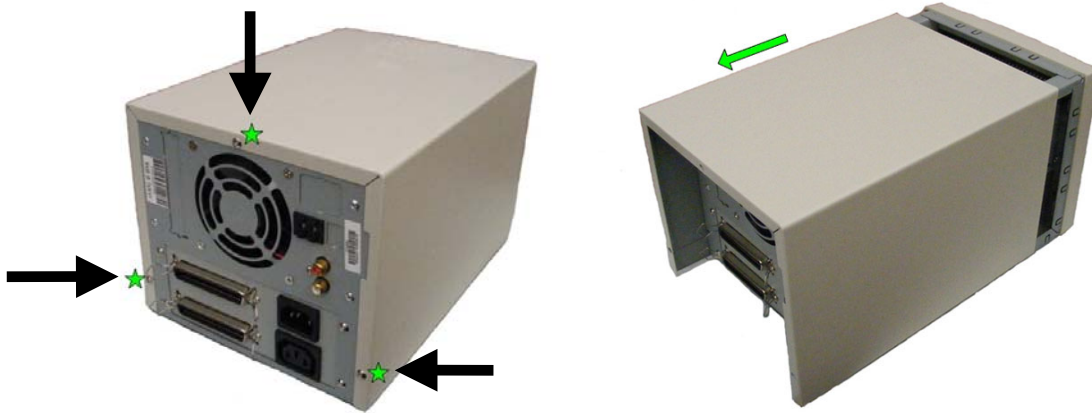
Illustration 1 : Old SCSI Tower – Remove power and SCSI cable



4. Remove Optical Drives from Old SCSI Tower

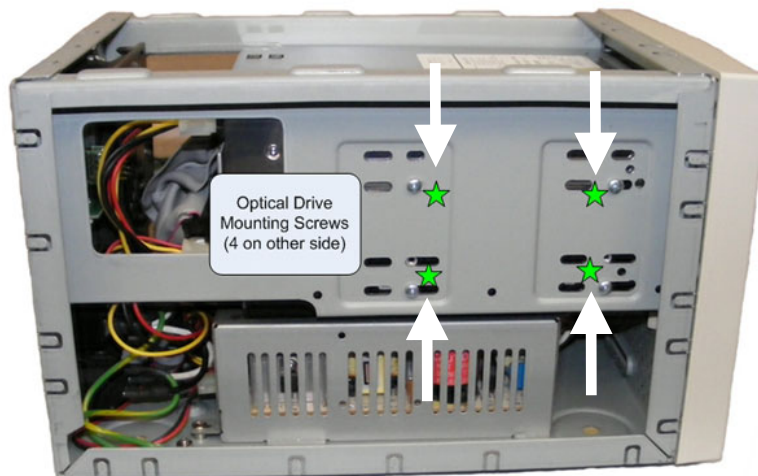
- 4.1. Remove the Cover from the Old SCSI Tower Chassis by removing the three (3) Phillips screws on the rear of the SCSI Tower Chassis and sliding the cover off. Use Dynema Gloves (Cut Resistant) to avoid injuries while removing the cover.

Illustration 2: Old SCSI Tower Chassis Cover Removal



- 4.2. Remove the Mounting Screws of both the drives (4 per drive) and set aside. You may need the mounting screws later.

Illustration 3: Old SCSI Tower Removal of Drives



- 4.3. Slide each drive forward and out the front of the chassis far enough to disconnect each Drive's power and ribbon cables.

Illustration 4: Old SCSI Tower Drives Removal



- 4.4. Completely remove **both** Drives from the Old SCSI Tower chassis. Place the drives in a static bag or on a static mat. Save the mounting screws.
- 4.5. Wearing a tested and proper fitting static strap, remove the SCSI to IDE Bridge Adapter Card from the rear of the DVD Drive.

NOTE: Remove the board carefully to avoid damage to the cable edge of the connector. The board is connected to the DVD-RAM only by the SCSI connector. There may be a tape holding the adapter board onto the drive. In that case, please remove the tape before removing the board.

Make sure jumper settings match the settings in Table 1, add or remove jumpers as necessary. Extra jumpers are provided in the FMI kit for use if needed. See Illustrations 5 to 7 for location of jumpers on the different board revisions. All revision boards are acceptable to use and jumper settings are same for each board, only location of jumpers on the board differ.

Table 1: Bridge Adapter Board (2368521) Jumper Setting

Jumper	Bridge Adapter Board (2368521) Jumper Setting	Function
JP1	In, out and out	Set to SCSI ID 4
JP2	In	Termination Enabled
JP3	out	Activity LED
JP4	In	Termination Power

Illustration 5: Bridge Adapter Rev 1.5

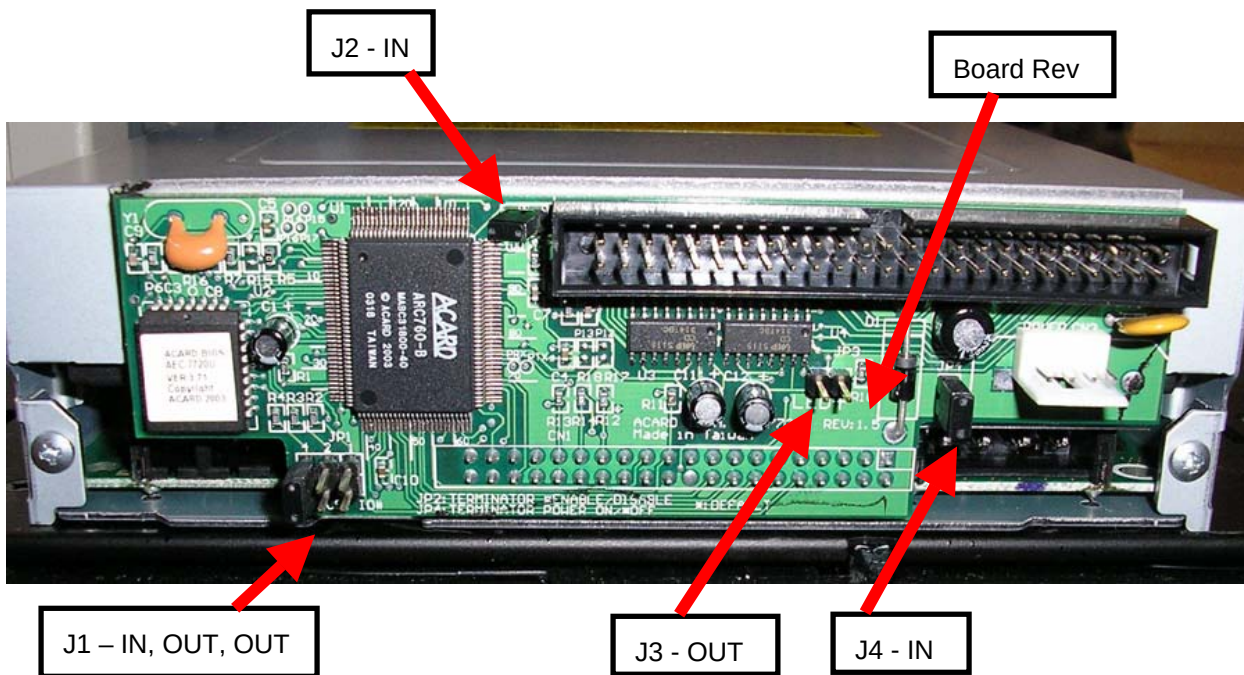


Illustration 6: Bridge Adapter Rev 2.0

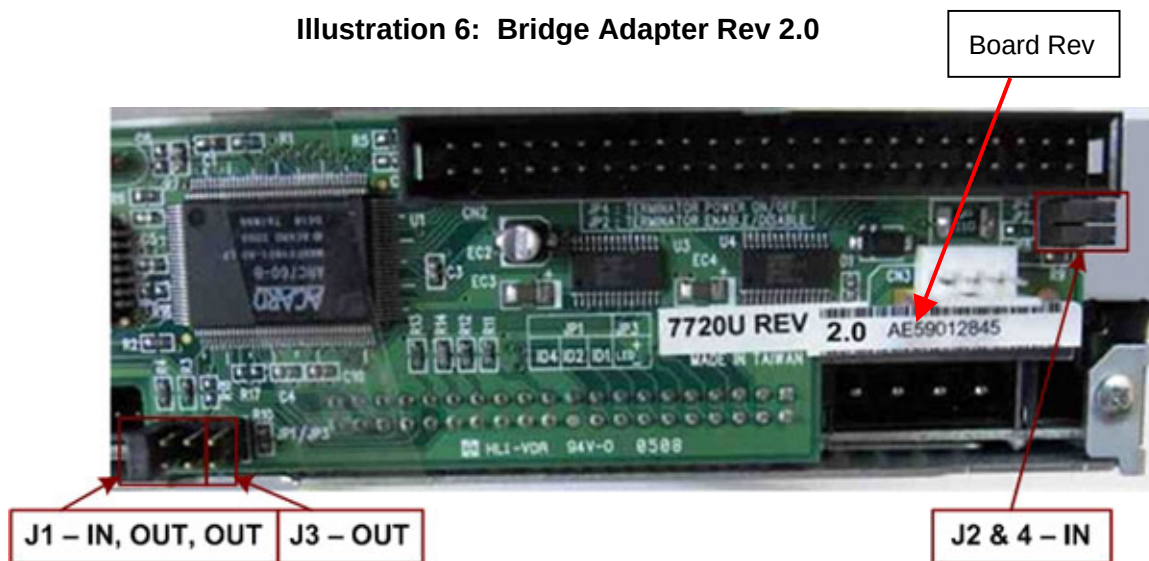
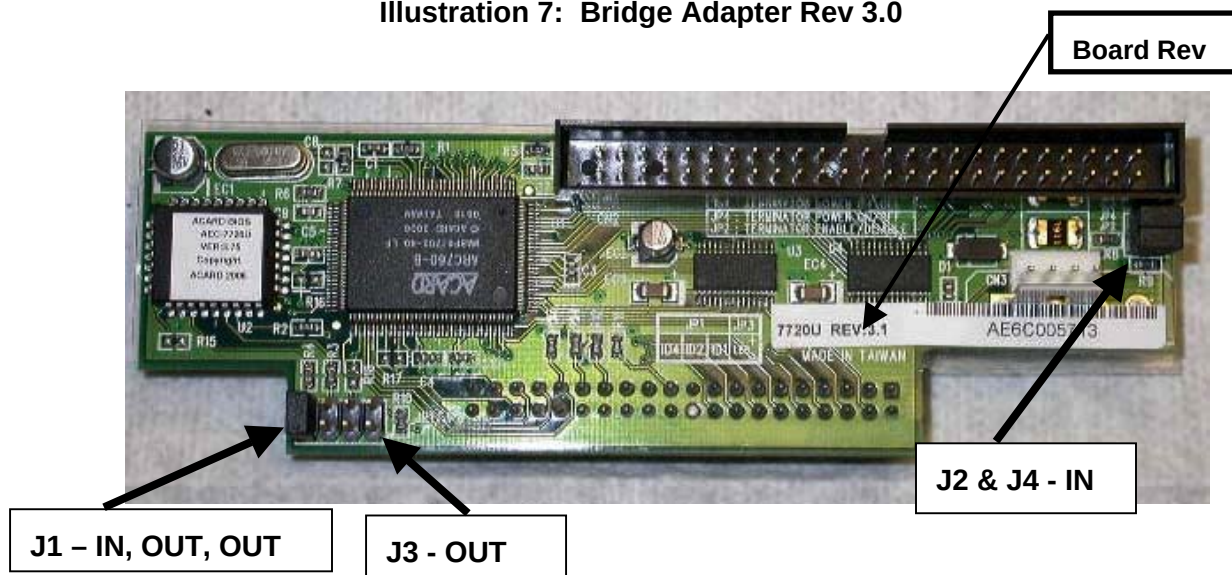


Illustration 7: Bridge Adapter Rev 3.0



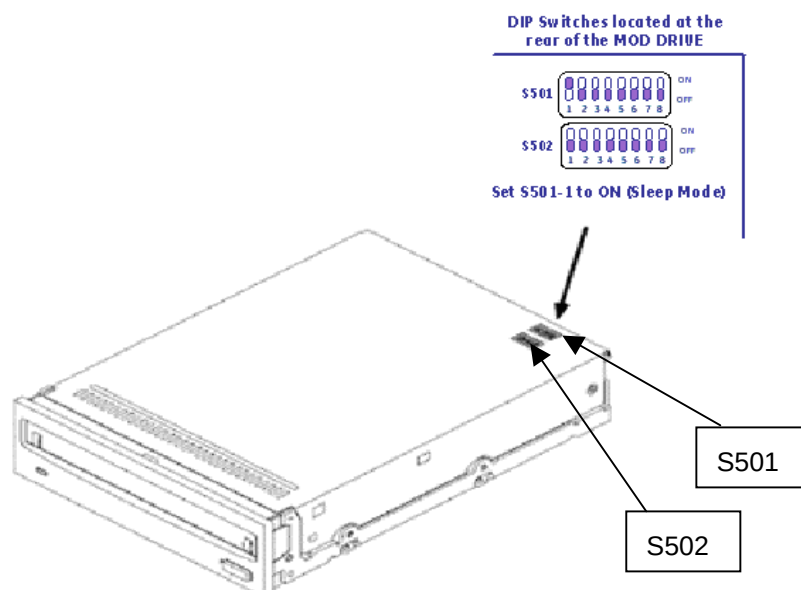
- 4.6. If the DVD Drive in the old tower (2368838) is not Panasonic Model Number SW-9576 (see label on top of the drive), dispose of the drive along with the old tower (see section 10).

If the DVD Drive in the old tower (2368838) is Panasonic SW-9576, this can be kept as spare part in a static bag.

- 4.7. Check positions of the Dip Switches on the old MOD Optical Drive. Make changes if necessary. Correct settings are shown below:

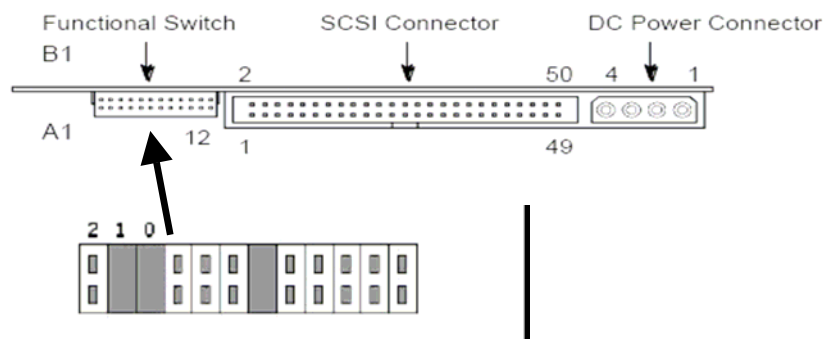
NOTE: Switch 1 on S501 DIP Switch is set to ON to allow MOD Sleep Mode. All other switches are set to OFF. Switches on S502 are all set OFF. As shown in the picture, Switch UP is ON and Switch Down is OFF.

Illustration 8: MOD Optical Drive DIP Switches



- 4.8. Check the jumpers in the Functional Switch of the MOD drive. Correct settings are shown below:

Illustration 9: MOD Optical Drive Jumpers

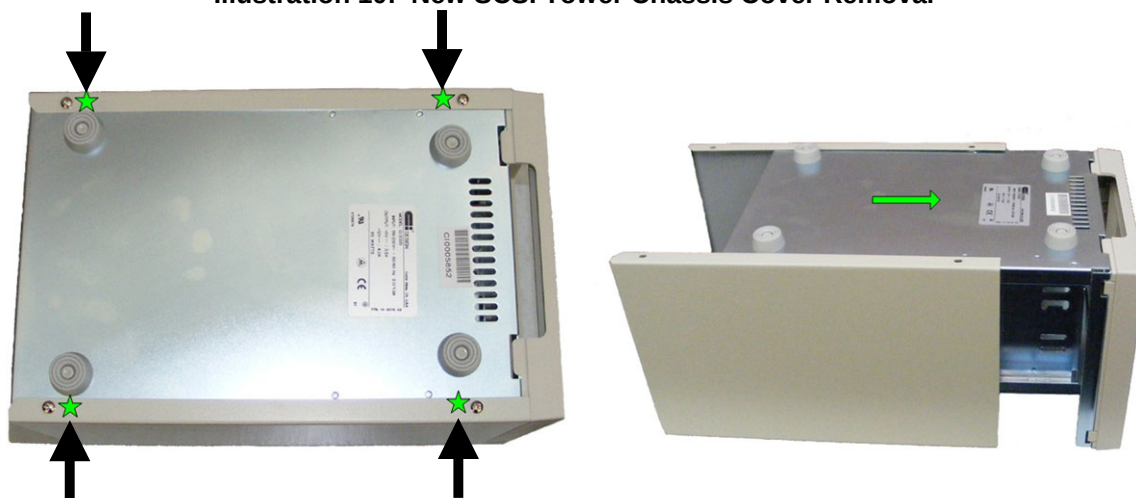


NOTE: Jumpers in location 2 (A2-B2) and location 3 (A3-B3) are to identify SCSI ID of 3. Jumper in location 7 (A7-B7) to Force Verify For Write Command

5. Install Optical Drives in New SCSI Tower

- 5.1. The SCSI Tower provided with the kit contains the new chassis with power supply with cables (5167410-10) and a new DVD Drive (2398863-2). Transfer the MOD Drive and the SCSI Adapter from the old 2368838 SCSI tower to the new tower per steps below.
- The MOD Optical Drive to be mounted on the top bay and the DVD on the bottom bay.
- 5.2. Remove four screws from the bottom of the New SCSI Tower Chassis.
- 5.3. Slide the cover to the rear and remove the enclosure. Use Dynema Gloves (Cut Resistant) to avoid injuries while removing the cover.

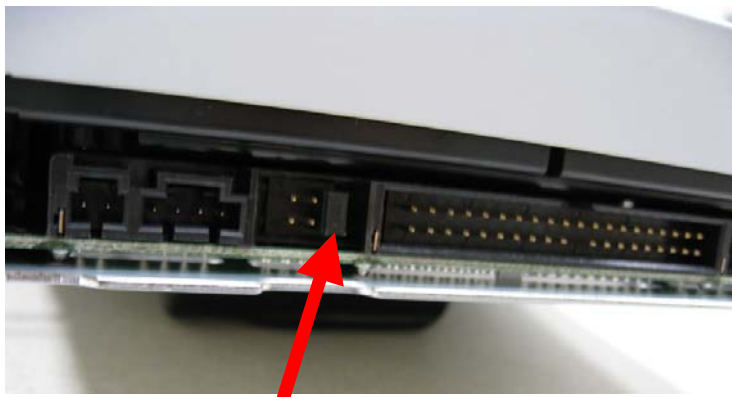
Illustration 10: New SCSI Tower Chassis Cover Removal



- 5.4. Remove the DVD Drive Mounting Screws and set aside.
- 5.5. Slide out and remove the DVD Drive.
- 5.6. Set the DVD configuration jumpers as shown below.

Illustration 11: DVD Drive Jumpers

NOTE: In the DVD drive, jumper the third location from the left as shown in the picture below.



Jumper location 3 from the left

- 5.7. Transfer the SCSI to IDE Bridge Adapter Card from the old DVD Drive to the new DVD Drive. Make sure that the Bridge Adapter board is properly seated and connected on the rear of the DVD Drive. *The board is connected to the DVD-RAM only by the SCSI connector.* The Bridge Adapter settings were done in step 4.5.

Illustration 12: Installation of Bridge Adapter Board

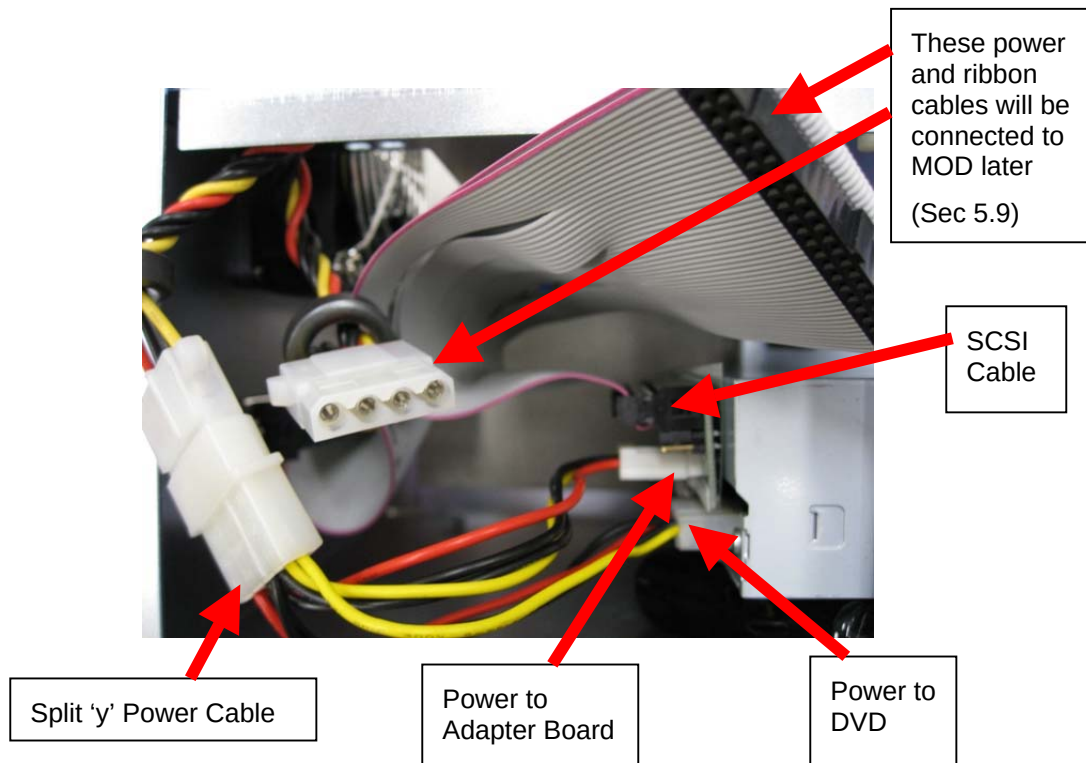


- 5.8. Lay the new tower on its side with front panel to the right side and the bottom closest to you. Slide in the DVD Drive into the new SCSI tower bottom bay part way.

Connect power cables to DVD Drive and Bridge Adapter board. Split 'y' power cables to DVD Drive and the Bridge Adapter board mounted on the back of the DVD.

Connect last SCSI connector on the ribbon cable to the SCSI connector on the Bridge Adapter Card.

Illustration 13: DVD Drive Install

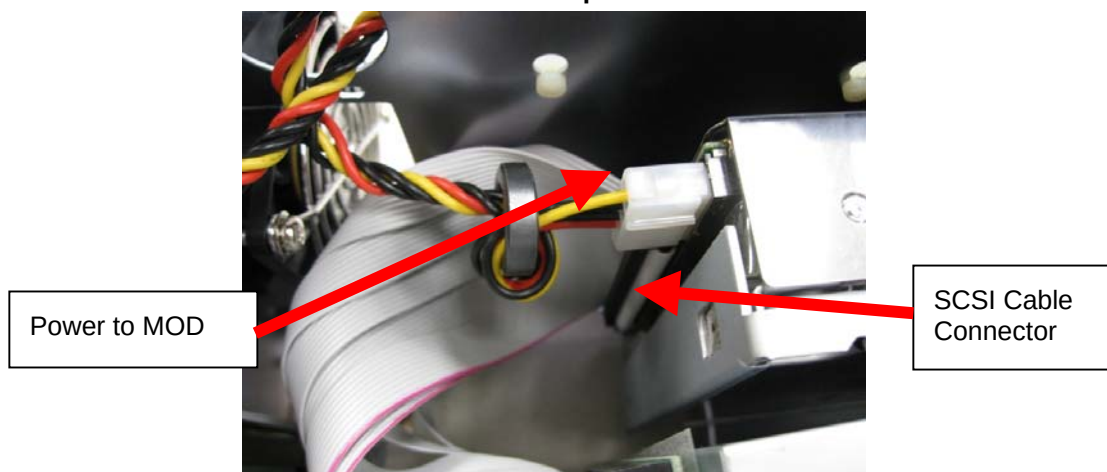


- 5.9 Take the MOD Optical Drive removed from the old SCSI Tower and slide it into the new SCSI Tower Chassis top bay part way and connect the middle ribbon cable connector to the MOD Drive.

Note: Make sure that the ribbon cable is oriented properly. The cable may need to be turned to get the right orientation.

Connect the 4 pin power connector from the SCSI Power Supply.

Illustration 14: MOD Optical Drive Install



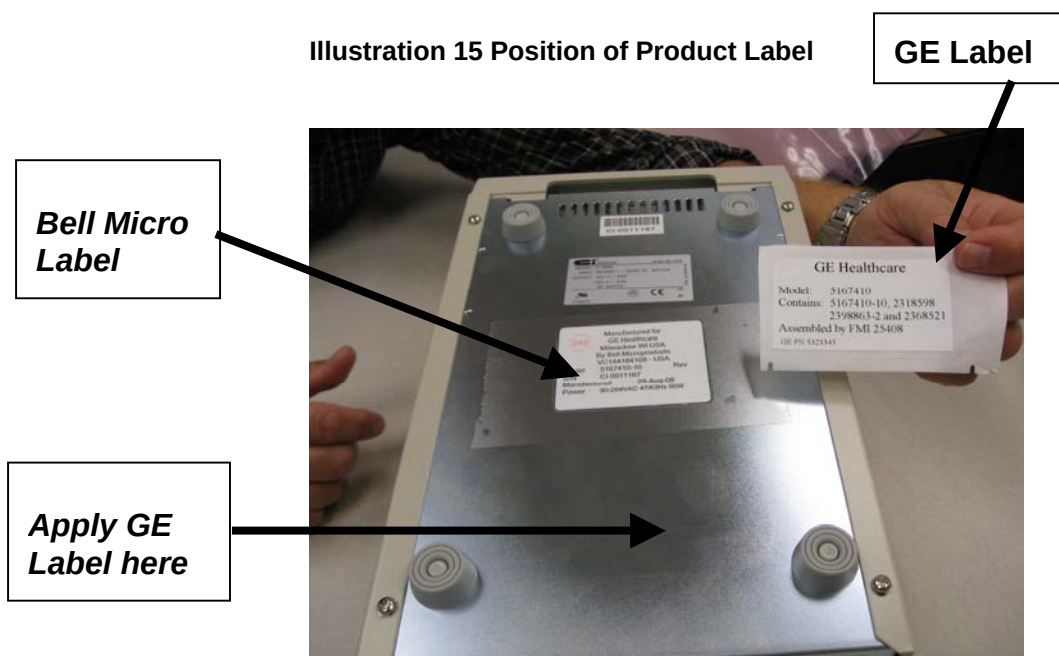
- 5.10. Continue to slide the DVD and MOD Optical Drives into the new SCSI Tower Chassis until they are flush with the front panel and mount the drives to the chassis using the Phillips Mounting Screws removed earlier.
- 5.11. Recheck all cable connections for proper seating.
Please make sure that the Bridge Adapter Board is seated properly to the DVD-RAM Drive.
- 5.12. Replace SCSI Tower Chassis Cover using the four (4) screws removed earlier. Use Dynema Gloves (Cut Resistant) to avoid injuries while replacing the cover.

New Label

Install product label 5323345(GE Label) provided with this FMI kit to the bottom of the tower.

Affix the label below the Bell Micro Label already on the bottom of the drive.
See Illustration 15 below.

Illustration 15 Position of Product Label



6. Connect New SCSI Tower to Operator Console

- 6.1. Connect the Operator Console SCSI Cable to the rear of the new SCSI Tower.
- 6.2. Connect the AC Power Cord to the rear of the new SCSI Tower.
- 6.3. Make sure that the Power Switch at the rear of the new SCSI Tower is turned ON.

7. Power On Operator Console

- 7.1. Apply power to the Operator Console and allow the system to fully boot up into Applications.
- 7.2. The Green Power LED on the front of the new SCSI Tower should be illuminated.

NOTE: If the system does not recognize SCSI devices, reboot the system.

8. Verify SCSI Tower Operation

8.1. DVD

- Insert formatted disk.
- Perform Save System Save and verify successful.

8.2. MOD

- Insert formatted MOD media.
- Save images to MOD and verify

NOTE: If the customer is not using MOD, there is no need to perform this test.

8.3. Remove all media from the SCSI Tower.

9. Disposal of old tower and/or old DVD Optical Drive

The old SCSI Tower and the old DVD-RAM should be disposed of properly. Please follow the local return process for the disposal.

10. FMI Closure

Close FMI per regional procedure.

Remove FMI Instructions from the customer site.